SPECTRAL GRAPH THEORY

Final Project

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1 Introduction

Spectral graph theory is a subsection of graph theory that leverages certain aspects of linear algebra to compliment the use of adjacency and Laplacian Matrices. It is our goal to introduce the basic concepts needed for spectral graph theory, and implement an investigation into the concepts, theorems and results from this section of math. Assumed knowledge is everything learnt in MA433 as of May 1st, 2024.

2 Prerequisites

Definition 2.1. A linear transformation is a function $T : \mathbb{R}^n \to \mathbb{R}^m$ that satisfy the following properties: 1. $T(x + y) = T(x) + T(y)$ 2. $T(ax) = aT(x)$.

Definition 2.2. An eigenvector is a vector that has its direction unchanged after a linear transformation. The linear transformation may scale the vector : $Av = \mu v$. μ is known as the eigenvalue.

Definition 2.3. The eigenvalues of a matrix A are the numbers λ such that $Ax = \lambda x$ has a nonzero solution vector; each such solution is an eigenvector associated with λ. The eigenvalues of a graph are the eigenvalues of its adjacency matrix A. These are the roots $\lambda_1, \ldots, \lambda_n$ of the characteristic polynomial $\phi(G; \lambda) = \det(\lambda I - A) = \prod_{i=1}^{n} (\lambda - \lambda_i)$.[1]

Eigenvalue & Eigenvector Example $A = \begin{bmatrix} 4 & 8 \\ 6 & 26 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{bmatrix} 4-\lambda & 8 \\ 6 & 26-\lambda \end{bmatrix}$$

$$(4-\lambda)(26-\lambda) - (8)(6) = 104 - 30\lambda + \lambda^2 - 48 = 0$$

$$\lambda^2 - 30\lambda + 56 = 0 \quad (\lambda - 28)(\lambda - 2) = 0$$

$$\lambda_1 = 28 \mid \lambda_2 = 2$$

$$N(A - \lambda_1 I) = \left[\begin{array}{cc|c} -24 & 8 & 0 \\ 6 & -2 & 0 \end{array}\right] \xrightarrow{R_2 + \frac{1}{4}R_1} \left[\begin{array}{cc|c} -24 & 8 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{cc|c} -3 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$-3x_1 + x_2 = 0 \quad x_2 = 3x_1 \quad \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$N\left(A-\lambda_2 I\right)=\left[\begin{array}{cc|c} 2 & 8 & 0 \\ 6 & 24 & 0 \end{array}\right] \xrightarrow{R_2-3R_1} \left[\begin{array}{cc|c} 2 & 8 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & 4 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$x_1+4x_2=0 \quad x_1=-4x_2 \quad \left[\begin{array}{l} -4 \\ 1 \end{array}\right]$$

3 Basic Concepts

We define the adjacency and Laplacian matrices. All graphs in this project should be assumed to be unweighted and undirected unless specified.

Definition 3.1. Let G be a loopless graph with vertex set $V(G)=\{v_1,\ldots,v_n\}$ and edge set $E(G)=\{e_1,\ldots,e_m\}$. The adjacency matrix of G, written $A(G)$, is the n-by-n matrix in which entry $a_{i,j}$ is the number of edges in G with endpoints $\{v_i,v_j\}$. The incidence matrix $M(G)$ is the n-by- m matrix in which entry $m_{i,j}$ is 1 if v_i is an endpoint of e_j and otherwise is 0 . If vertex v is an endpoint of edge e, then v and e are incident. The degree of vertex v (in a loopless graph) is the number of incident edges. [1]

3.1 Properties

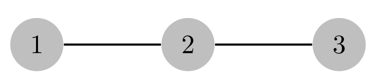
- The matrix is symmetric for undirected graphs: $A=A^T$.
- The diagonal elements are zero if there are no loops in the graph.
- The sum of the elements in any row (or column) of A is the degree of the corresponding vertex.

3.2 Example

Consider a simple undirected graph with vertices $1,2,3$, and edges $\{1,2\}$, $\{2,3\}$. The adjacency matrix A is:

$$A=\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

3.3 Graph Drawing



Definition 3.2. A *Laplacian Matrix* is defined as the determinant matrix - the adjacency matrix of a graph G. Using this matrix, we can derive eigenvectors and eigenvalues that unveil coloring and connectivity properties of G.

3.4 Properties

- L is symmetric and positive semi-definite.
- Each entry l_{ij} is -1 if there is an edge between vertices i and j, and 0 otherwise.
- Each diagonal entry l_{ii} equals the degree of vertex i.
- The sum of the elements in any row or column of L is zero.
- The smallest eigenvalue of L is 0, and the corresponding eigenvector is the one vector.

3.5 Example

Using the same graph as in the adjacency matrix example, the degree matrix D and the Laplacian matrix L are given by:

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Theorem 3.3. If $f(x) = x^T Ax$, where A is a real symmetric matrix, then f attains its maximum and minimum over unit vectors x at eigenvectors of A, where it equals the corresponding eigenvalues.

Proof: The function f is continuous in $x_1, \ldots, x_n$. For constrained optimization, we use Lagrangian multipliers. Given the constraint $x^T x = 1$, we let $g(x) = x^T x - 1$. Forming $L(x, \lambda) = f(x) - \lambda g(x)$, the extreme values occur where all partial derivatives of L are 0. With respect to λ, this yields $x^T x = 1$. Let ∇ denote the vector of partial derivatives with respect to $x_1, \ldots, x_n$. We compute $\nabla L(x, \lambda) = \nabla f(x) - \lambda g(x) = 2Ax - 2\lambda x$. The statement $\nabla f(x) = 2Ax$ uses the symmetry of A. We have $\nabla L = 0$ precisely when $Ax = \lambda x$ which requires x to be an eigenvector of A for eigenvalue λ. This yields $f(x) = x^T Ax = \lambda x^T x = \lambda$. [1]

Theorem 3.4. (Spectral Theorem) A real symmetric $n \times n$ matrix has real eigenvalues and n orthonormal eigenvectors.

Proof:
We use induction on n. The claim is trivial for $n = 1$; consider $n > 1$. Let v_n be the eigenvector maximizing $x^T Ax$. Let W be the orthogonal complement of the space spanned by v_n; it has dimension $n - 1$. If $w \in W$, then ${v^T}_n Aw = w^T Av_n = \lambda_n w^T v_n = 0$. Hence $Aw \in W$. Viewing multiplication by A as a mapping f_A we have $f_A : W \to W$.

Let S be a matrix whose columns are the vectors of an orthonormal basis of $\mathbb{R}^n$ with v_n as the last column. Since the basis is orthonormal, $S^{-1} = S^T$. The matrix for f_A with respect to this basis is $S^T AS$ is 0, except for λ_n in the last position. Furthermore, the matrix is symmetric. Hence its first $n-1$ rows and columns form the matrix A' for f_A on W with respect to this basis.
By the induction hypothesis, A' has orthonormal eigenvectors $v_1, \ldots, v_{n-1}$, with real eigenvalues. Using S, we convert these back into real eigenvectors for A. They ahve the same real eigenvalues, and they form an orthonormal set.

Next we consider polynomial functions of a matrix. Viewed as members of $\mathbb{R}^{n^2}$, the matrices $I, A, A^2, \ldots, A^{n^2}$ cannot be independentm, since there are n^2+1 of them. Using an equation of linear dependence, we obtain a polynomial p such that $p(A)$ is the zero matrix. The characteristic polynomial itself suffices. This holds for all A, but again we consider only real symmetric matrices. [1]

Theorem 3.5. The diameter of a graph G is less than the number of distinct eigenvalues of G.

proof:
Let A be the adjacency matrix; A satisfies a polynomial of degree r if and only if some linear combination of $A^0, \ldots, A^\top$ is 0 . Since the number of distinct eigenvalues is the degree of the minimum polynomial, we need only show that $A^0, \ldots, A^k$ are linearly independent when $k \leq \operatorname{diam}(G)$.

It suffices to show for $k \leq \operatorname{diam}(G)$ that A^k is not a linear combination of $A^0, \ldots, A^{k-1}$. Choose $v_i, v_j \in V(G)$ such that $d(v_i, v_j) = k$. By counting walks, we have $A^k_{i,j} \neq 0$ but $A^!_{i,j} = 0$ for $t < k$. Therefore, A^k is not a linear combination of the smaller powers.

Since the Spectral Theorem guarantees real eigenvalues, we can index our eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_n$. We also refer to λ_1 and λ_n as $\lambda_m(G)$ and $\lambda_.$ (G) [1]

3.6 Spectral Properties and Graph Structure

Spectral graph theory reveals deep connections between the eigenvalues of a graph's matrices and its structural properties such as connectivity, diameter, and expansion.

3.6.1 Connectivity and Expansion

The eigenvalues of the Laplacian Matrix provide crucial insights into a graph's connectivity. Particularly, the second smallest eigenvalue (algebraic connectivity or Fiedler value), can tell a lot about a graph. A graph is connected iff this value is greater than zero. The size of this eigenvalue also provides a measure of the graph's robustness to cuts. Higher values suggests a more connected graph.

Theorem 3.6. The algebraic connectivity of a graph is greater than zero if and only if the graph is connected.

3.7 Spectral Diameter

The diameter of a graph is the shortest path between the two furthest nodes. It can be estimated using spectral methods. A general rule is that the diameter is inversely related to the spectral gap, which is the largest and second largest eigenvalues of the adjacency matrix.

Theorem 3.7. The diameter of a graph G is bounded above by a function of the spectral gap of G's adjacency matrix.

This relationship provides a computational method to estimate the diameter of large graphs where direct computation would be impractical, and it is especially useful in the analysis of large networks such as those found in social media or biology.

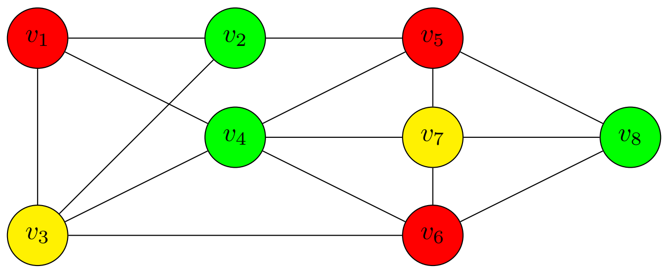
These connections between spectral properties and graph structure not only provide a deeper understanding of graph theory but also enable practical applications across various fields, from network analysis to machine learning and beyond.

4 Graph Colorings

4.1 Chromatic Number

Definition 4.1. The *Chromatic Number*, $\chi(\mathrm{G})$, of a graph G is the minimum number of color classes required to color a graph so that no two neighboring vertices are the same color.

Chromatic Number Ex.



Since the largest complete subgraph of G is K_3,$\chi(\mathrm{G}) = 3$. We find a relationship between $\chi(\mathrm{G})$ and the largest eigenvalue of G's adjacency matrix.

4.2 Spectral Radius

Definition 4.2. The *spectral radius* of a graph G is the largest eigenvalue λ of the adjancency matrix A.

$$\mathrm{J}_{m_i}(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_i & 1 \\ 0 & 0 & \cdots & 0 & \lambda_i \end{bmatrix} \in \mathbf{C}^{m_i \times m_i}, 1 \leq i \leq s$$

[2]

Using the spectral radius, we can conclude that the chromatic number is can be no greater than $\lambda + 1$ thanks to Wilf's Theorem.

4.3 Upper Bound of Chromatic Number

Theorem 4.3. (Wilf [1967]) For every graph $G, \chi(G) \leq 1 + \lambda_{\max}(G)$.[1]

Lemma 1. For every graph $G, \delta(G) \leq \frac{2e(G)}{n(G)} \leq \lambda_{\max}(G) \leq \Delta(G)$. largest coordinate value in x. Then $\lambda \leq \Delta(G)$ follows from

$$\lambda x_j = (Ax)_j = \sum_{v_i \in N(v_j)} x_i \leq d(v_j)\, x_j \leq \Delta(G) x_j.$$

Lemma 2. If G' is an induced subgraph of G, then

$$\lambda_{\min}(G) \leq \lambda_{\min}(G') \leq \lambda_{\max}(G') \leq \lambda_{\max}(G)$$

Proof: If $\chi(G) = k$, then we can successively delete vertices without reducing the chromatic number until we obtain a subgraph H such that $\chi(H - v) = k - 1$ for all $v \in V(H)$. Since we know that $\delta(H) \geq k - 1$, and H is an induced subgraph of G, Lemma 1 and then Lemma 2 yield.

$$k \leq 1 + \delta(H) \leq 1 + \lambda_{\max}(H) \leq 1 + \lambda_{\max}(G).$$

Sylvester's Law of Inertia yields a lower bound on the number of bicliques needed to decompose a graph. Because stars are bicliques and every subgraph of a star is a star, the number of bicliques needed is at most the vertex cover number $\beta(G) = n(G) - \alpha(G)$. Erdós conjectured that equality almost always holds, but this remains open. Graphs with special structure may have efficient partitions using other bicliques.[1]

4.3.1 Applying Wilf's to Chromatic Number Example

Continuing the graph coloring example from 4.1, we can find the upper bound for the chromatic number by finding the maximum eigenvalue of the adjacency matrix A.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$\lambda_i = [-2.32812303, -1.68574835, -1.16308111, -0.765753, -0.03625806, 0.97269517, 2.05080028, 3.95546811]$

According to Wilf's, χ(G) $\leq \lambda_{max} + 1 = 4.95546811$.

Since we know χ(G) $= 3$, the theorem holds.

4.4 Eigenvalue Properties

Theorem 4.4. (Perron-Frobenius) If the adjacency matrix A of a graph G is non-negative, then the eigenvector corresponding to its spectral radius, λ, must also be non-negative.

Proof. The proof is quite geometric and intuitive. Look at the sphere $x_1^2 + \ldots + x_n^2 = 1$ and intersect it with the space $\{x_1 \geq 0, \ldots, x_n \geq 0\}$ which is a quadrant for $n = 2$ and octant for $n = 3$. This gives a closed, bounded set X. The matrix A defines a map $T(v) = Av/|Av|$

on X because the entries of the matrix are nonnegative. Because they are positive, TX is contained in the interior of X. This map is a contraction, there exists $0 < k < 1$ such that $d(Tx, Ty) \leq kd(x, y)$ where d is the geodesic sphere distance. Such a map has a unique fixed point v by Banach's fixed point theorem. This is the eigenvector $Av = \lambda v$ we were looking for. We have seen now that on X, there is only one eigenvector. Every other eigenvector $Aw = \mu w$ must have a coordinate entry which is negative. Write $|w|$ for the vector with coordinates $|w_j|$. The computation

$$|\mu||w|_i = |\mu w_i| = \left|\sum_j A_{ij} w_j\right| \leq \sum_j |A_{ij}| \, |w_j| = \sum_j A_{ij} \, |w_j| = (A|w|)_i$$

shows that $|\mu|L \leq \lambda L$ because $(A|w|)$ is a vector with length smaller than λL, where L is the length of w. From $|\mu|L \leq \lambda L$ with nonzero L we get $|\mu| \leq \lambda$. The first " $\leq$ " which appears in the displayed formula is however an inequality for some i if one of the coordinate entries is negative. Having established $|\mu| < \lambda$ the proof is finished. [3]

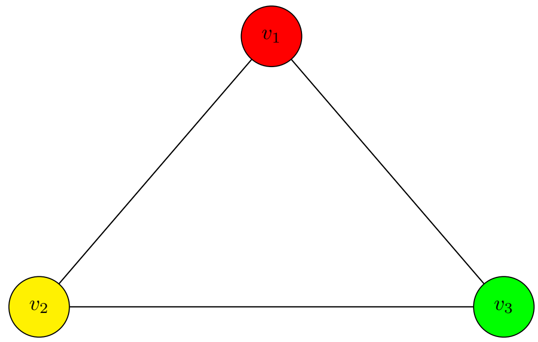
4.5 Regular Graphs

Theorem 4.5. (Hoffman [1963]) A graph G is regular and connected if and only if J is a linear combination of powers of $A(G)$.[1]

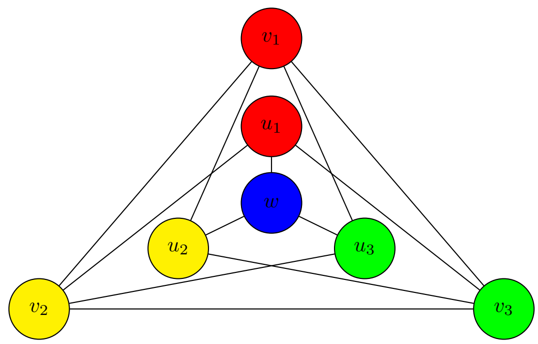
Proof: Suffciency. If J can be so expressed, then for each i, j we have $\left(A^k\right)_{ij} \neq 0$ for some $k \geq 0$, which requires a v_i, v_j-walk of length k. Hence G is connected. For regularity, consider the matrices JA and AJ. The i, j th position of AJ is $d\left(v_i\right)$ (constant on rows), and the i, j th position of JA is $d\left(v_j\right)$ (constant on columns). Since J is a linear combination of powers of A, each of which commutes with A, we have $JA = AJ$. Thus the i, j th position is both $d\left(v_i\right)$ and $d\left(v_j\right)$ and the graph is regular.
Necessity. Since G is k-regular, k is an eigenvalue, and the minimum polynomial is $\psi(G; \lambda) = (\lambda - k) g(\lambda)$ for some polynomial g. Since $\psi(G; A) = 0$, we have $\mathrm{Ag}(A) = kg(A)$. Hence each column of $g(A)$ is an eigenvector of A with eigenvalue k. Since G is regular and connected, each such eigenvector is a multiple of $\mathbf{1}_n$. Hence the columns of $g(A)$ are constant. However, $g(A)$ is a lin. symmetric. Hence the columns are equal and $g(A)$ is a multiple of J. [1]

4.6 Mycielski's Construction

Let G be a graph and let G' be obtained from G by Mycielski's construction: beginning with G having vertex set $\{v_1, v_2, \ldots, v_n\}$, add vertices $U = \{u_1, u_2, \ldots, u_n\}$ and one more vertex w. Add edges to make u_i adjacent to $N(v_i)$, all of the neighbors of v_i, and finally let $N(w) = U$.



G



G'

If each u_i is adjacent to all neighbors of v_i, u_i and v_i should be the same color to color G' most efficiently. With this, w will be adjacent to vertices of every color of G. Therefore, w must be a new color and $\chi(G') = \chi(G) + 1$. [4]

5 Bipartite Graphs

The adjacency matrix A of a bipartite graph has a bipartite structure, meaning that it can be described as:

$$A = \begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix},$$

where B is a matrix representing the connections between the two partitions. This structure leads to a symmetric spectrum about zero:

- If λ is an eigenvalue of A, so is $-\lambda$. This includes both positive and negative versions of non-zero eigenvalues, as well as zero if it appears as an eigenvalue.
- Eigenvalues come in $\pm\lambda$ pairs unless $\lambda = 0$, which relates to the graph's bipartite nature.

5.1 Laplacian Matrix

The Laplacian matrix $L = D - A$, where D is the degree matrix, also exhibits particular spectral properties:

- The smallest eigenvalue of L is 0, and the graph is connected if and only if the eigenvalue 0 has multiplicity one.
- For connected bipartite graphs, the spectrum of L can be particularly informative about the graph's expansion properties and its resistance to cuts.

Theorem 5.1. Let $G = (V, E)$ be a bipartite graph with bipartition (V_1, V_2). The spectrum of the adjacency matrix A of G is symmetric about zero.

Suppose G is bipartite, and can be divided into two disjoint sets V_1 and V_2 such that no two vertices within the same set are adjacent. Consider the adjacency matrix A of G. By the structure of G, A can be partitioned as:

$$A = \begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix}$$

where B is a matrix describing the edges between vertices in V_1 and V_2.

Suppose λ is an eigenvalue of A with eigenvector $\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}$, where $\mathbf{x}_1$ corresponds to vertices in V_1 and $\mathbf{x}_2$ to vertices in V_2. The eigenvalue equation $A\mathbf{x} = \lambda\mathbf{x}$ can be expressed as:

$$\begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}$$

This leads to the system of equations:

$$\begin{cases} B\mathbf{x}_2 = \lambda\mathbf{x}_1 \\ B^T\mathbf{x}_1 = \lambda\mathbf{x}_2 \end{cases}$$

Now consider the vector $\mathbf{y} = \begin{bmatrix} \mathbf{x}_1 \\ -\mathbf{x}_2 \end{bmatrix}$. Applying A to $\mathbf{y}$, we have:

$$A\mathbf{y} = \begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ -\mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} -B\mathbf{x}_2 \\ B^T\mathbf{x}_1 \end{bmatrix} = \begin{bmatrix} -\lambda\mathbf{x}_1 \\ \lambda\mathbf{x}_2 \end{bmatrix} = -\lambda \begin{bmatrix} \mathbf{x}_1 \\ -\mathbf{x}_2 \end{bmatrix} = -\lambda\mathbf{y}$$

Thus, $\mathbf{y}$ is an eigenvector of A corresponding to the eigenvalue $-\lambda$.

Hence, for every eigenvalue λ of A, there exists an eigenvalue $-\lambda$. This proves the spectrum around A is symmetric.

5.2 Algebraic Connectivity

The second smallest eigenvalue of the Laplacian, known as the algebraic connectivity or Fiedler value, indicates the graph's robustness against disconnection. In bipartite graphs, this value can be used to assess the ease of separating the graph into its two partitions.

5.3 Cheeger's Inequality

For bipartite graphs, Cheeger's Inequality provides a bound on the graph's edge expansion, which is a measure of how difficult it is to partition the graph into two large components connected by few edges:

$$h(G) \geq \frac{\lambda_2}{2},$$

where $h(G)$ is the Cheeger constant and λ_2 is the second smallest eigenvalue of the Laplacian.

Proof:

Let f be any non-constant function from the vertices of G to real numbers, orthogonal to the constant function (in the sense that the sum over all vertices of f times the degree of the vertex is zero). Normalize f so that $\sum_{v \in V} d_v f(v)^2 = 1$, where d_v is the degree of vertex v.

The Rayleigh quotient for f and the Laplacian L is given by:

$$R(f) = \frac{\sum_{\{u,v\} \in E} (f(u) - f(v))^2}{\sum_{v \in V} d_v f(v)^2}$$

Given the setup, we know that $\lambda_2 \leq R(f)$ by the definition of eigenvalues and the variational principle.

Next, define the subset S of vertices by:

$$S = \{v \in V : f(v) > 0\}$$

The boundary $|\partial S|$ (the number of edges crossing from S to $V \setminus S$) can be lower bounded in terms of f as:

$$|\partial S| \geq \sum_{\{u,v\} \in \partial S} |f(u) - f(v)| \geq \sum_{\{u,v\} \in \partial S} |f(u)| + |f(v)|$$

Using the Cauchy-Schwarz inequality, we further get:

$$|\partial S| \geq \frac{(\sum_{\{u,v\} \in \partial S} |f(u) - f(v)|)^2}{\sum_{\{u,v\} \in \partial S} 1}$$

By definition, $h(G) \leq \frac{|\partial S|}{\min(|S|, |V \setminus S|)}$. Rearranging terms and considering f as a function approximating the characteristic function of S, we see:

$$h(G) \leq \frac{(\sum_{\{u,v\} \in E} |f(u) - f(v)|)^2}{\sum_{v \in V} d_v f(v)^2 \cdot \min(|S|, |V \setminus S|)}$$

Since $\sum_{v \in V} d_v f(v)^2 = 1$ and $\sum_{\{u,v\} \in E} (f(u) - f(v))^2 \leq \lambda_2$, we find:

$$h(G) \geq \frac{\lambda_2}{2}$$

This concludes the proof of Cheeger's Inequality.

5.4 Spectral Graph Theory in Bipartite Graphs

Spectral graph theory provides the tools to analyze the properties of graphs using matrices associated with them. In bipartite graphs, these spectral properties offer important interpretations and can be used to solve problems.

5.4.1 Interlacing Theorems

Spectral interlacing theorems provide a way to understand how the eigenvalues of a graph's matrix change as the graph is modified. For bipartite graphs, these theorems can help predict the effects of operations like vertex or edge deletion:

Theorem 5.2. If a graph G' is a subgraph of a bipartite graph G, then the eigenvalues of G' interlace the eigenvalues of G.

This theorem implies that removing vertices or edges from G cannot increase the maximum eigenvalue or decrease the minimum eigenvalue of its adjacency matrix.

5.4.2 Spectral Bipartiteness Measure

The concept of spectral bipartiteness measures how close a graph is to being bipartite, using spectral methods. It is defined in terms of the eigenvalues of the normalized Laplacian matrix $\mathcal{L}$ of the graph:

$$\beta(G) = \sqrt{2(1 - \lambda_2)}, \tag{1}$$

where λ_2 is the second smallest eigenvalue of $\mathcal{L}$. For a perfectly bipartite graph, $\beta(G)$ achieves its maximum value.

5.4.3 Spectral Partitioning and Clustering

Spectral clustering techniques can leverage the properties of bipartite graphs to identify community structures or clusters within the graph. The Fiedler vector (eigenvector corresponding to the second smallest eigenvalue of the Laplacian) often reveals a natural bipartition of the graph, which is particularly straightforward in bipartite graphs due to their spectral properties:

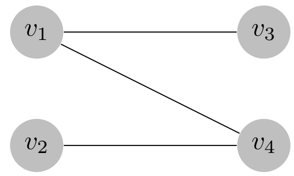
- For a bipartite graph, the Fiedler vector can perfectly separate the two partitions.
- This method is highly effective in applications such as image segmentation, data clustering, and social network analysis.

5.4.4 Example of a Bipartite Graph

Consider a simple bipartite graph with partition sets $V_1 = \{v_1, v_2\}$ and $V_2 = \{v_3, v_4\}$, and edges $\{v_1, v_3\}$, $\{v_1, v_4\}$, and $\{v_2, v_4\}$.

Adjacency Matrix The adjacency matrix for this graph is:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$



5.5 Analysis of a Bipartite Graph's Adjacency Matrix

Given the adjacency matrix A of a bipartite graph:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

with the partition matrices B and B^T:

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

We calculate BB^T and B^TB to find:

$$BB^T = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad B^TB = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

The eigenvalues of BB^T and B^TB lead us to the characteristic polynomial:

$$\lambda^2 - 3\lambda + 1 = 0$$

Solving this, we obtain eigenvalues:

$$\lambda = \frac{3 \pm \sqrt{5}}{2}$$

Thus, the eigenvalues of A are:

$$\pm\sqrt{\frac{3+\sqrt{5}}{2}}, \quad \pm\sqrt{\frac{3-\sqrt{5}}{2}}$$

These eigenvalues reflect the bipartiteness of the graph. The symmetry about zero in the spectrum indicates that for every positive eigenvalue there is an equal and opposite eigenvalue. This means that there is a balance between a bipartite structure. This fundamental characterstic of bipartite graphs impacts their spectral and structural properties.

6 Applications

When on social media, the information received and passed out by any given user will mainly be contained in the web of accounts that are followed. More often than not, people will be friends with those who are connected to many of the same accounts, allowing the content from those accounts to be spread rapidly through these communities.

Using the normalized Laplacian Matrix, we can derive the normalized eigenvectors and eigenvalues. Using these values and an algorithmic approach, we can find the optimal partition in any graph. From this, we can visualize the clustering of a given graph and see partitions between webs of tightly connected data points. This concept can be applied to real life networks, like social media, to visualize the groups in which people communicate and surround themselves online.

To gather data, Eugen gathered his own data from Instagram, analyzing who he follows, who follows him, and which accounts Eugen and his followers share in following. From this, we compiled a data set to run through our machine learning algorithm. We were able to visualize the social networks that Eugen is a part of on Instagram, and thanks to spectral clustering we could partition his network from those that his followers may be a part of.

6.0.1 Normalized Laplacian

The normalized Laplacian is defined as:

$$\mathcal{L} = D^{-1/2} L D^{-1/2} = I - D^{-1/2} A D^{-1/2}$$

where I is the identity matrix. This matrix is particularly useful because it handles graphs with varying degrees better by normalizing the influence of each node according to its degree.

6.0.2 Spectral Decomposition

The normalized Laplacian matrix $\mathcal{L}$ is decomposed into its eigenvalues and eigenvectors:

$$\mathcal{L}\mathbf{u}_i = \lambda_i \mathbf{u}_i$$

where λ_i are the eigenvalues and $\mathbf{u}_i$ are the corresponding eigenvectors. The eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq ... \leq \lambda_n$ are real and non-negative due to the positive semi-definiteness of $\mathcal{L}$.

6.0.3 Using Eigenvectors for Clustering

To cluster the graph:

1. Select the first k eigenvectors $\mathbf{u}_1, \mathbf{u}_2, ..., \mathbf{u}_k$ corresponding to the smallest k eigenvalues.
2. Form the matrix $U \in \mathbb{R}^{n \times k}$ with these eigenvectors as columns.
3. Normalize each row of U to have unit length, resulting in matrix U'.
4. Treat each row of U' as a point in $\mathbb{R}^k$, and cluster these points into k clusters via k-means or another clustering algorithm.

[5]

This method effectively partitions the graph based on the connectivity and can reveal communities within networks such as social media platforms.

6.1 Real Life Analysis on Instagram

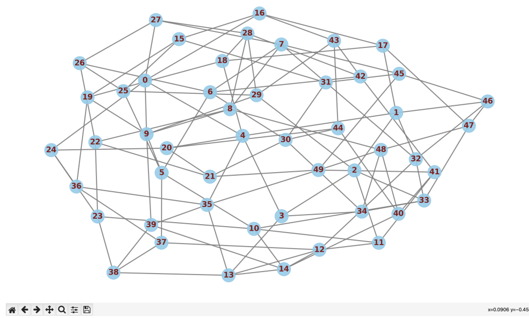


Figure 1: Insta friends

Above in Figure 1 is a subsection of 50 friends from Instagram and their connections with each other. It is a tightly woven interconnection with different communities; however, we can find a bottleneck to partition the data into groups.

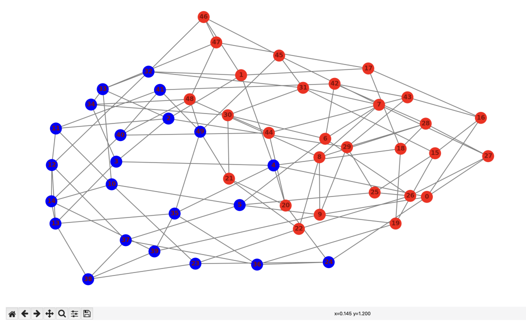


Figure 2: Bottlenecks

Figure 2 shows this bottleneck and the two main communities in the following. Doing some investigations within the Instagram, we found that this was a BU community and a non-BU community.

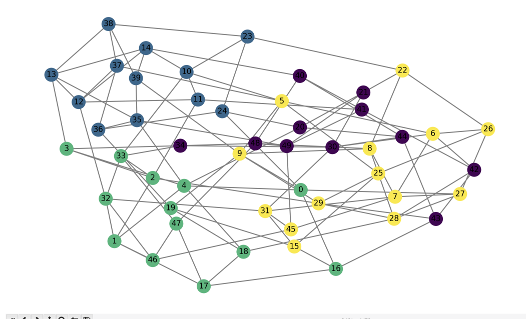


Figure 3: 4 clusters

Figure 3 incorporates a technique of spectral clustering, where we chose 4 clusters arbitrarily to demonstrate that sub-communities can be formed within communities. For a comprehensive spectral clustering algorithm, a high degree of machine learning and trial and error is needed.

This methodology goes to show how people can become surrounded by like-minded individuals and become sheltered to information presented outside their sub-communities. These sub-communities could be for any number of interests, from similar interests in sports, politics, celebrity culture, etc.

On a broader scale, this phenomenon of bottlenecks and sub-communities within the two communities in a bottleneck results in what some refer to as an ”echo chamber”, particularly on social media. With the rise of political discourse on social media, polarization within our main parties in the U.S. has also sharply increased. Social media allows people to surround themselves with people saying what they want to hear, and it allows for a blocking out of opposing views and hampers healthy discourse on issues.

The analysis done is cited below on a github repository where there is a python file and a json file that details the graph. For privacy reasons, the names are randomized and no person is directly mentioned in the json file. [6]

References

[1] D. B. West, "Introduction to graph theory, second edition," 2000.

[2] Wikipedia, "Spectral radius."

[3] O. Knill, "Math 19b: Linear algebra with probability," 2011.

[4] J. Duque-Rosero, "Ma433 homework 6," 2024.

[5] https://www.geeksforgeeks.org/ml-spectral clustering/, "Spectral clustering."

[6] E. Nesbakken, "Github repository: Ma433 project," 2024.